Experiment 18:- Write a Program to Implement Email - Spam Naive Bayes Classifier

In [1]:
```
import pandas as pd
df = pd.read_csv('spam.csv')
df.head(20)
df.shape
```

Out[1]:
```
(5572, 2)
```

In [2]:
```
df.groupby('Category').describe()
```

Out[2]:

	Message			
	count	unique	top	freq
Category				
ham	4825	4516	Sorry, I'll call later	30
spam	747	641	Please call our customer service representativ...	4

In [3]:
```
df['spam']= df['Category'].apply( lambda x: 1 if x == 'spam' else 0)
df.head()
```

Out[3]:

	Category	Message	spam
0	ham	Go until jurong point, crazy.. Available only ...	0
1	ham	Ok lar... Joking wif u oni...	0
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	1
3	ham	U dun say so early hor... U c already then say...	0
4	ham	Nah I don't think he goes to usf, he lives aro...	0

In [4]:
```
from sklearn.model_selection import train_test_split
x_train , x_test , y_train , y_test = train_test_split(df.Message , df.spam , tes
```

In [5]:
```
# Here CountVectorizer converts text into numbers by counting
# how many times each word appears
# This numerical representations is called as Bag-of-Words Representation
from sklearn.feature_extraction.text import CountVectorizer
v = CountVectorizer()
x_train_count = v.fit_transform(x_train.values)
x_train_count.toarray()[:]
#print(v.get_feature_names())
x_train_count.shape
```

Out[5]:
```
(4457, 7768)
```

In [6]:
```
from sklearn.naive_bayes import MultinomialNB
model = MultinomialNB()
model.fit(x_train_count, y_train)
```

Out[6]:

▾ MultinomialNB

▸ Parameters

▸ Fitted attributes

In [7]:
```
x_test_count = v.transform(x_test)
model.score(x_test_count , y_test)
```

Out[7]:
```
0.9856502242152466
```